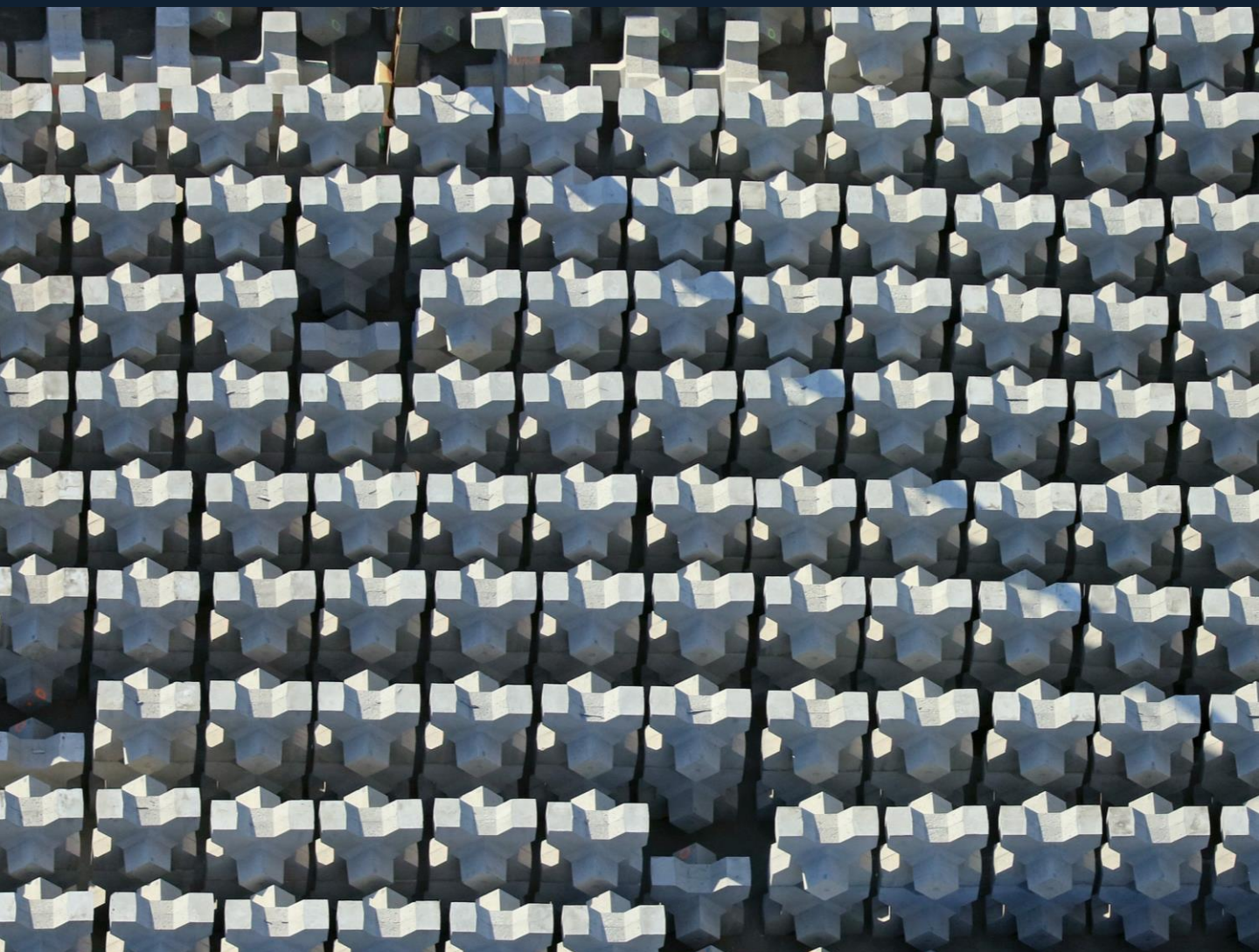


Health & Safety

Standard Logistics and storage



Reference

Company	Document Type	Department	N°	Version	Langue
BYTP	REF	H&S	2217	A	EN

Version	Date	Write by	Checked by	Approved	Reason for modification
A	01-11-2023	O.BRZEZINSKI	I.LELIEVRE	L.KNOLL	First issue

The logistics and storage standards presented in this document serve as a practical guide to define the means to be planned and implemented to effectively manage flows and storage.

The elements of this document are completed by the elements presented in the reference documents **BYTP-REF-H&S-2212 [Vehicles / pedestrian interaction]** for the organization of vehicles/pedestrian interfaces, **BYTP-REF-H&S-2210 [Lifting Standard]** for the organization of lifting and **BYTP-REF-H&S-2218 [Ergonomics Standard]** for handling aids.

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1. Management of logistics flows of materials/equipment

Logistics is a key factor in the efficient planning, production, financial management and safety success of the various construction sites

Benefits of logistics optimization:

- Enhanced Health and Safety on the project
- Waste reduction
- Minimal on-site storage requirements

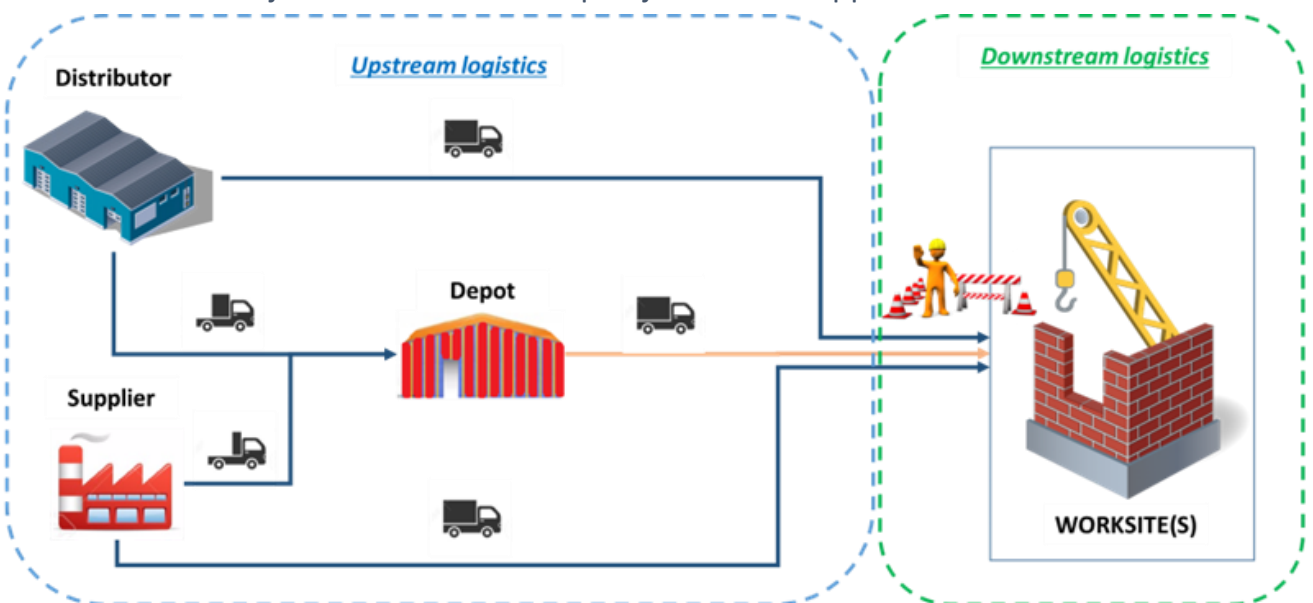


During the consultation / site preparation phase:

- Select appropriate on-site storage areas, rear bases or provisional waiting areas
- Foresee and plan goods journeys, both external and internal on site (considering them as well for the selection of suppliers)
- Designate a logistics manager for operations/sites that require it
- Anticipate appropriate lifting and handling equipment, based on the packages arriving at the project and the delivery areas (at height, in shafts, etc.)
- Plan for packaging and means of gripping
- Consider access roads and traffic management on site (site layout)

During the construction phase:

- Implement the planned logistical choices from the preparation phase and make any necessary adaptation(s)
- Store the elements on site in designated, clean and unobstructed areas
- Ensure the recruitment of skilled/trained personnel in various roles: traffic marshal, banksman/signaller, gantry crane operator, lift operator, etc.
- Facilitate on-the-job deliveries and adopt a just-in-time approach



2. Loading/unloading protocol

The loading/unloading protocol incorporates useful information for assessing the risks generated by the operation as well as the prevention and safety measures to be observed at each stages of the operation.

Before deliveries, the Works Manager sends the updated loading/unloading protocol to the suppliers . It may be attached to the order (Interface with the Purchasing Department if applicable), and should be returned signed by the supplier before the delivery takes place.

For example in France, a template for the loading/unloading protocol is usually available as an appendix to the safety coordination documents or the Health & Safety Plan for interventions with third parties.

This document is used for preparing safe deliveries.

The essential information that appears there (non-exhaustive list):

- Address, site opening hours
- Name of "receiver" (internally designated to receive deliveries)
- Type of operations
- Vehicle specifications
- Type of goods
- Identification of risks and preventive measures
- Safety instructions (access, accident, etc.)

This information is communicated to suppliers before delivery.

BOUYGUES TRAVAUX PUBLICS

PROTOCOLE DE SECURITE Chargement/déchargement

Chantier : Viaduc A9 Lez-Lironde

Nature et objet de l'opération																	
☐ Chargement :	☐ Déchargement :																
Documents remis ou commentés par BYTP RF	Observations particulières																
☐ Plans de circulation détaillés : • Plots et axes de circulation • Zones de stockage • Zones de retournement	Contacter : A. THOMAS (Bouygues) 0661616029 A. THOMAS (Chef de Chantier) 0661378836 A. BERNARD (Securité) 0692201710																
☐ Consignes d'urgence en cas d'accident, incident																	
Consignes générales de sécurité BP obligatoires pour intervenir sur le site :     Autres : • Respecter la signalisation routière sur le site (limitation de vitesse...) • Prendre aux véhicules sortants du chantier. • Stationner sur les axes autorisés, en dehors des plots de circulation. • Signaler le stationnement au l'arrêt par l'usage des feux de freinage. • Un véhicule doit être correctement à l'arrêt avant toute intervention. • Communiquer l'opération de chargement / déchargement avec l'accord du responsable de l'opération. • Utiliser le matériel mais à disposition par les opérateurs. Ce matériel doit être conforme à la réglementation en vigueur. • Interdiction d'utiliser les engins du site. • Respecter l'environnement et le classement du chargement.																	
<table border="1"> <thead> <tr> <th colspan="2">BYTP RF</th> <th colspan="2">Entreprise extérieure</th> </tr> </thead> <tbody> <tr> <td colspan="2">Nom/Prénom :</td> <td colspan="2">Entreprise :</td> </tr> <tr> <td>Date :</td> <td></td> <td>Date :</td> <td></td> </tr> <tr> <td>Visa :</td> <td></td> <td>Visa :</td> <td></td> </tr> </tbody> </table>		BYTP RF		Entreprise extérieure		Nom/Prénom :		Entreprise :		Date :		Date :		Visa :		Visa :	
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BOUYGUES TRAVAUX PUBLICS

PROTOCOLE DE SECURITE Chargement/déchargement

Chantier : Viaduc A9 Lez-Lironde



3. Unloading from the ground

To avoid climbing onto truck decks or high access platforms:

- Use permanent slings/chains preferably
- Anticipate using hooking devices which can be operated from the ground (reinforcement, precast parts, etc.) for equipment transport

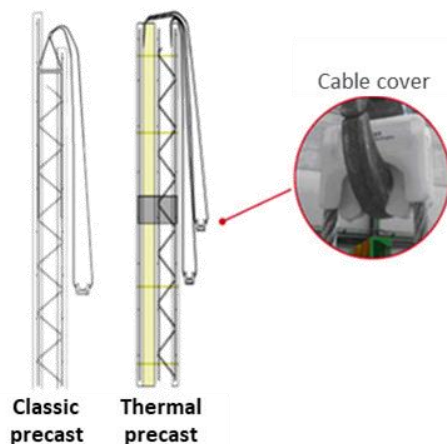
Bring the hooking point close to the worker
and not the other way round



River caissons



Reinforcement cages



4. Access from an unloading dock

Use of an unloading dock (or more than one to cover the entire platform) to access the packages/equipment/materials to be unloaded:



Fixed docks



Usable LIRP
to sling



Truck
access



Mobile docks



5. Site containers



The condition of site containers reflects the level of tidiness and organization maintained at the construction sites.

- They should be organized and kept clean
- Defective material can be isolated in a dedicated to this purpose
- Storing materials on the rooftop is not permitted
- They should be equipped with permanent lifting chains that are in proper condition Has a water extinguisher + additive
- They should be equipped with necessary firefighting equipment
- If chemicals are stored (excluding flammable chemicals to be stored outdoor on retention pallet), the container must be adequately ventilated using both high and low grilles
- Auxiliary heaters are prohibited.

Periodic checks should be conducted on all materials stored within the container to ensure their current validity

Smoking is prohibited in the containers.



Container power supply

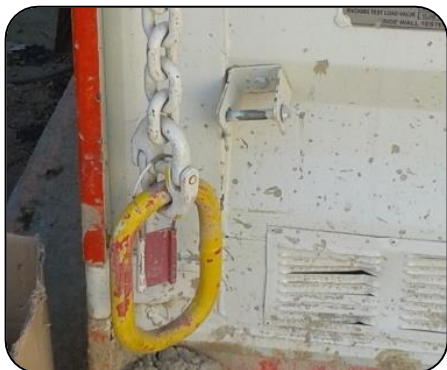
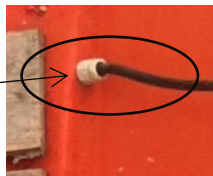
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Grounding + cable gland

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Verification by an approved organization

Cable gland



Hanging lifting slings/chains with chain retention device during transport



Equipment container equipped with permanent lifting chains

6. Storage areas

Materialization of storage areas:

- Define and identify reserved areas
- Identify the different storage areas
- Use geotextile on a clean surface to prevent soiling of steels during storage



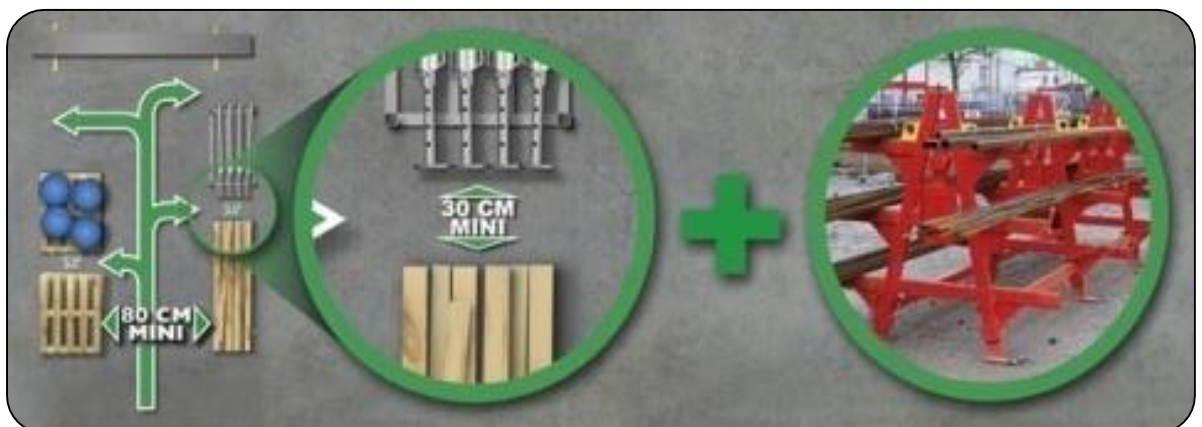
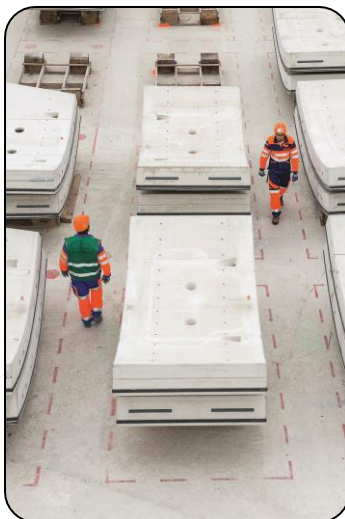
Circulation in storage areas and access to hanging points:

- Maintain circulation aisles inside storage areas (30 cm for packages and 80 cm for circulation)
- Guarantee secure access to all storage points



Height of slinging points = 1.50m

Bring the hooking point close to the worker and not the other way round



7. Storage racks

Storage racks allow:

- Optimization of storage areas
- Facilitation of material handling
- Ensuring the storage stability of non-self-stabilizing elements (beams, stringers, etc.)

Storage racks that can be moved by crane should have validated anchor points and a SWL displayed



Rack à mannequins



Rack for precast parts



Rack for rolling bars



Rack for precast walls (side stabilizations required)

8. Handling assistance systems

Assistance with handling = reduction of arduousness



Trolley for props



Paver trolley



Necessary equipment available on rolling cart

Refer to the Ergonomics Standard

9. Storage baskets

The 5-sided closed storage baskets allow:

- Optimization of storage spaces
- Facilitation of handling (compartmentalized baskets)
- Ensuring compliant lifting (identified anchoring points + SWL)

It is necessary to respect the SWL of the baskets



Baskets closed on 5 sides and adapted to the load to be lifted



Compartmental basket

When the basket is not closed on 5 sides, make the load monolithic by strapping the stored material.



Baskets not closed on 5 sides = mandatory strapping



The containers of the baskets do not exceed the height of the uprights

Refer to the Lifting Standard

Post storage baskets

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Clean working areas and reduced manual handling

10. FIBC (Flexible Intermediate Bulk Container)

- Big bags must be lifted with handling baskets
- Prohibit the delivery of bulk reinforcements in big bags



Loading height not to exceed: 2/3



No handling of scrap metal in big bags



- self-emptying
- Do not reuse single-use big bags



Refer to the Lifting Standard